

Case Study No.: 4

Case Study Title

ATM Card PIN Verification System Using JavaScript Functions

Theory

An **ATM Card PIN Verification System** is a simple security application that verifies whether the entered ATM PIN is a **palindrome**. A palindrome is a sequence of digits or characters that reads the same from left to right and right to left. Although real banking systems do not use palindrome checking for authentication, this case study demonstrates the practical implementation of JavaScript functions, scope, closures, and exception handling in a real-world scenario.

The application accepts a PIN from the user, reverses the entered PIN using a function, compares it with the original PIN, and displays whether it is a palindrome. If the PIN is a palindrome, a special security message is displayed; otherwise, the user is informed that the PIN is not a palindrome.

Function Declaration

A function declaration defines a reusable block of code using the **function** keyword. It is used to perform operations such as cleaning or validating the entered PIN.

Example:

- `function cleanPin(pin){`
- `return pin.trim();`
- `}`

Function Expression

A function expression stores a function inside a variable. It is commonly used for reversing the entered PIN.

Example:

- `const reversePin = function(pin){`
- `return pin.split('').reverse().join('');`
- `};`

Arrow Function

Arrow functions provide a shorter syntax for writing functions and are useful for simple comparisons.

Example:

```
const isPalindrome = (original, reversed) => original ===  
  reversed;
```

Scope

Scope defines where variables are accessible within a program.

- **Global Scope** – Variables declared outside functions are accessible throughout the program.
 - **Local Scope** – Variables declared inside functions can only be accessed within that function.
 - **Block Scope** – Variables declared using `let` and `const` inside a block `{ }` are limited to that block.
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Closure

A **closure** is a function that remembers variables from its outer function even after the outer function has finished execution. In this case study, a closure is used to count the number of PIN verification attempts.

Working of the System

1. The user enters an ATM PIN.
2. JavaScript validates the entered PIN.
3. The PIN is reversed using a function expression.
4. An arrow function compares the original and reversed PIN.

5. If both values are equal, the PIN is identified as a palindrome.
 6. A special security message is displayed for palindrome PINs.
 7. A closure keeps track of the number of verification attempts.
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Features

- Accepts ATM PIN from the user.
 - Reverses the PIN using a JavaScript function.
 - Checks whether the PIN is a palindrome.
 - Demonstrates Function Declaration.
 - Demonstrates Function Expression.
 - Demonstrates Arrow Function.
 - Demonstrates Global and Local Scope.
 - Demonstrates Closure.
 - Displays user-friendly success and error messages.
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Applications

- Banking System Simulations
 - ATM Security Demonstrations
 - JavaScript Learning Projects
 - Programming Logic Development
 - Educational Applications
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Advantages

- Improves understanding of JavaScript functions.
- Demonstrates real-world implementation of function concepts.
- Enhances logical thinking.
- Provides hands-on experience with closures and scope.
- Interactive and user-friendly interface.
- Easy to understand and maintain.

1. Case Study Program Code:

index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ATM Portal - Palindrome PIN Verification</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:'Segoe UI',Tahoma,Geneva,Verdana,sans-serif;
}

body{
    height:100vh;
    display:flex;
    justify-content:center;
    align-items:center;
    background:linear-gradient(135deg,#0f172a,#1e3a8a,#3b82f6);
}

.atm-container{
    width:400px;
```

```
background:#fff;
border-radius:20px;
padding:35px;
box-shadow:0 20px 45px rgba(0,0,0,.25);
}

.header{
  text-align:center;
  margin-bottom:30px;
}

.user-badge{
  color:#64748b;
  text-transform:uppercase;
  letter-spacing:2px;
  font-size:12px;
}

.header h2{
  color:#2563eb;
  margin:10px 0;
}

.user-name{
  color:#111827;
  font-size:18px;
  font-weight:600;
  line-height:1.6;
}

.input-group{
  margin:25px 0;
}

label{
  display:block;
  margin-bottom:10px;
  color:#475569;
  font-size:13px;
  font-weight:600;
}
```

```
}

input{
    width:100%;
    padding:15px;
    border:2px solid #dbeafe;
    border-radius:12px;
    background:#f8fafc;
    font-size:22px;
    text-align:center;
    letter-spacing:5px;
    transition:.3s;
}

input:focus{
    outline:none;
    border-color:#2563eb;
    box-shadow:0 0 12px rgba(37,99,235,.25);
}

button{
    width:100%;
    padding:15px;
    border:none;
    border-radius:12px;
    background:linear-gradient(135deg,#2563eb,#1d4ed8);
    color:white;
    font-size:16px;
    font-weight:bold;
    cursor:pointer;
    transition:.3s;
}

button:hover{
    transform:translateY(-2px);
    box-shadow:0 12px 25px rgba(37,99,235,.35);
}

.screen-display{
    margin-top:25px;
}
```

```
    min-height:80px;
    padding:18px;
    border-radius:12px;
    display:flex;
    justify-content:center;
    align-items:center;
    text-align:center;
    background:#f8fafc;
    border:2px solid #e2e8f0;
    color:#64748b;
    line-height:1.5;
    transition:.3s;
}

.special-msg{
    background:#dcfce7;
    color:#166534;
    border-color:#22c55e;
    font-weight:bold;
}

.error-msg{
    background:#fee2e2;
    color:#b91c1c;
    border-color:#ef4444;
    font-weight:bold;
}

.footer{
    margin-top:20px;
    text-align:center;
    color:#94a3b8;
    font-size:12px;
}

</style>
</head>

<body>
```



```
<div class="atm-container">

<div class="header">

<div class="user-badge">
Secure ATM Portal
</div>

<h2>&img alt="ATM icon" data-bbox="161 254 181 268"/> ATM LOGIN</h2>

<div class="user-name">
Radhika Chaudhari<br>
PRN : 24070521150
</div>

</div>

<div class="input-group">

<label>ENTER YOUR 4-DIGIT PIN</label>

<input
type="text"
id="pinInput"
maxlength="4"
placeholder="Enter PIN"
autocomplete="off"
inputmode="numeric">

</div>

<button id="verifyBtn">
Verify PIN
</button>

<div class="screen-display" id="atmScreen">
Welcome! Please enter your 4-digit PIN.
</div>

<div class="footer">
```

```
Palindrome PIN Verification System
</div>

</div>

<script>

// Function Declaration
function reversePIN(pin) {
    return pin.split('').reverse().join('');
}

// Function Expression
const isPalindrome = function(pin) {
    return pin === reversePIN(pin);
};

// Closure
function createATMVerifier(){

    const successMessage =
        "✅ PIN Verified Successfully! Welcome to Secure ATM Banking.";

    // Arrow Function
    const verifyATMCard = (enteredPin)=>{

        if(enteredPin.length !== 4 || isNaN(enteredPin)){

            return{
                status:"error",
                message:"❌ Please enter a valid 4-digit numerical PIN."
            };

        }

        if(isPalindrome(enteredPin)){

            return{
                status:"special",
                message:successMessage
            };
        }
    };
}
```

```

        };

    }

    return{

        status:"error",
        message:"❌ Access Denied! PIN is NOT a Palindrome."

    };

};

return verifyATMCard;
}

const atmSystem = createATMVerifier();

const pinInput = document.getElementById("pinInput");
const verifyBtn = document.getElementById("verifyBtn");
const atmScreen = document.getElementById("atmScreen");

// Allow only digits
pinInput.addEventListener("input", ()=>{

    pinInput.value = pinInput.value.replace(/[^0-9]/g, "");

});

// Verify Button
verifyBtn.addEventListener("click", ()=>{

    const result = atmSystem(pinInput.value);

    atmScreen.className = "screen-display";

    if(result.status==="special"){

        atmScreen.classList.add("special-msg");
    }

```

```
        alert(
"⚠ SECURITY WARNING!\n\n" +
"The PIN you entered is a PALINDROME.\n\n" +
"Palindrome PINs are considered weak because they are easier to
guess.\n\n" +
"Please change your PIN after logging in for better security."
        );

    }

    else{

        atmScreen.classList.add("error-msg");

    }

    atmScreen.innerHTML = result.message;

});

// Press Enter to Verify
pinInput.addEventListener("keypress", (e)=>{

    if(e.key==="Enter"){

        verifyBtn.click();

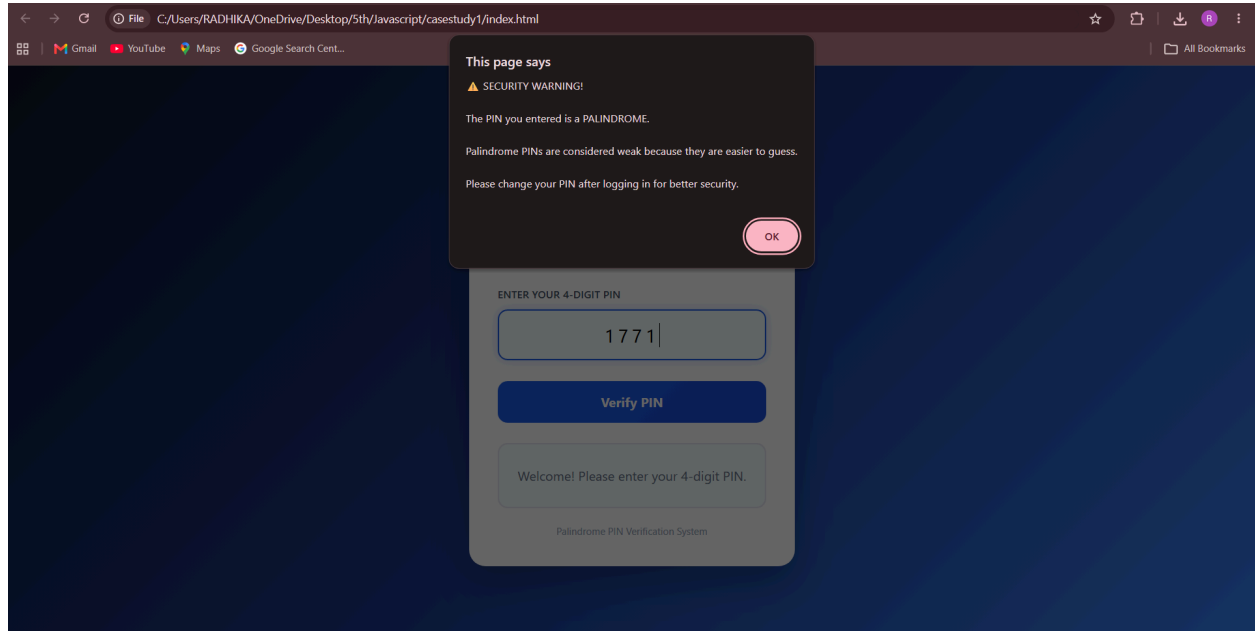
    }

});

</script>

</body>
</html>
```

Output:



Result/Conclusion:

The **ATM Card PIN Verification System** was successfully implemented using **HTML5, CSS3, and JavaScript**. The application accepts an ATM PIN, reverses it using a function expression, and checks whether it is a palindrome using an arrow function. The project demonstrates the concepts of **function declaration, function expression, arrow functions, variable scope, closures, and JavaScript-based validation**. The system accurately identifies palindrome and non-palindrome PINs while providing an interactive and user-friendly interface, thereby enhancing the understanding of JavaScript functions and their real-world applications.

